

## CSCE 670 INFORMATION RETRIEVAL · MODULE 11 · LECTURE NOTES

# Generative Retrieval

*When the answer is written rather than found.*

## MODULE OUTCOMES

- Explain the architecture from the retrieval side.
- Explain what attribution requires.
- Explain how evaluation changes.
- Explain the effects on the documents and their authors.
- Build and evaluate an answering system.

## 1 The shift

### RETURNING AN ANSWER INSTEAD OF A LIST MOVES THE JUDGEMENT FROM THE USER TO THE SYSTEM

**A result list lets the user judge relevance for themselves** — they see the titles, the source domains, and the snippets, and they choose which to trust. **An answer has already chosen on their behalf.**

**Which is better when the system is right and considerably worse when it is wrong** — because **the user has lost both the alternatives and the cues they were using to judge** (Module 01 §3's relevance-as-relation point: the user was doing work the system cannot do).

**And it changes the failure mode entirely: a bad result list is visibly bad and a fluent wrong answer is not** (CSCE 638 Module 12 §3's mechanism), which means the user cannot apply the scepticism that a poor list would have triggered.

**So attribution is not a feature but the mechanism that restores the user's ability to judge — which is why §2 treats it as a requirement rather than a nicety**, and why an answering system without working citations is a worse product than a list.

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QUERY UNDERSTANDING (Module 09)
    rewrite, decompose, resolve against the session

RETRIEVAL (Modules 03, 07, 08)
    hybrid, then rerank. THIS IS STILL THE
    BOTTLENECK -- a generator cannot answer from
    passages it was not given.

SELECTION
    how many passages, which ones, and in what order.
    Position effects within the context are real, so
    put the best first (CSCE 638 Module 11
    section 4's lost-in-the-middle row).

GENERATION
    answer FROM the passages, with citations, and
    abstain when they do not suffice

AND THE RETRIEVAL HALF IS WHERE THE QUALITY IS. The
generation step is largely solved and the retrieval
step is not, which is the opposite of where the
attention usually goes.

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The retrieval half is where the quality is — and it is the half that gets the least attention in practice, because the prompt is the most visible and most easily modified component (CSCE 638 Module 11 §2's same observation). **Framing this as a retrieval problem rather than a generation problem is this course's contribution to the topic**, and it is the right framing: everything in Modules 02 through 09 applies unchanged, and the generation step is a presentation layer over it.

## 2 Attribution

**A CITATION MUST SUPPORT THE SPECIFIC CLAIM IT IS ATTACHED TO, WHICH IS A STRONGER REQUIREMENT THAN IT LOOKS**

The weak version of attribution is that the cited document is topically related to the answer — which is what a system produces by default if it retrieves passages and then generates freely while listing its sources, **and it is nearly worthless** as a verification aid.

The actual requirement is that the cited passage entails the specific sentence it is attached to — which has to be checked by reading, and frequently fails even when the retrieval was good and the answer is broadly correct.

And the characteristic failure is a plausible claim with a citation to a document that does not actually say it — which is worse than no citation at all, because the citation confers credibility the claim has not earned and discourages the checking it appears to invite.

So measure attribution directly: sample sentences from generated answers, read the cited passage, and record whether it supports the claim — which is slow, unavoidable, and the only honest measurement available (CSCE 638 Module 11 §4's fourth evaluation).

- **Generate per-sentence rather than per-answer citations**, so that **each individual claim is traceable** rather than the answer as a whole being attributed to a set of sources.
- **Instruct the model to quote before it paraphrases, which makes the support checkable mechanically** — a quoted span either appears in the cited passage or it does not, and that check is a string search.
- **Verify with an entailment model** and drop or flag the unsupported sentences — **which is a second model and is worth it where the stakes are real**, and which is how the better production systems handle this.
- **And present the passage itself, not only a link to it**, so that the user can check without navigating away — **which is the cheapest improvement available** and the most effective.
- **While accepting that none of these eliminate the problem** (CSCE 638 Module 12 §3) — **it is structural**, and a system that claims to have solved it has not measured it. **Presenting the supporting passage inline is the cheapest and most effective of these, because it restores the user's judgement**, which is §1's argument.

### 3 Evaluation

- **Retrieval recall at the passage level**, measured separately — **because it bounds everything downstream and is the thing classical IR evaluation already measures well** (Module 05), so there is no excuse for omitting it.
- **Answer correctness, established by reading** — against a reference answer where one exists, and by judgement where one does not (which is most interesting cases, and is CSCE 638 Module 10 §3's unsolved problem).
- **The attribution rate (§2)** — **the fraction of claims genuinely supported by the citations attached to them**, sampled and read.
- **The abstention rate on unanswerable questions**, measured by **deliberately asking some that your corpus cannot answer** (CSCE 638 Module 11 §4) — which takes an hour to set up and is almost never done.
- **And the user-side measures: did they have to check the sources? did they find the answer sufficient? did they reformulate? Reformulation and checking behaviour are the strongest available signals that an answer did not satisfy — and they are free**, arriving in the logs (Module 09 §3's reformulation signal).

### 4 Consequences

#### AN ANSWER THAT SATISFIES THE USER REMOVES THE VISIT THE DOCUMENT'S AUTHOR DEPENDED ON

The web's documents were written under an implicit arrangement: search engines send visitors, and the visit is what the author receives in exchange for having produced the content — through advertising, subscriptions, reputation, or sales.

An answered query breaks that arrangement — the user is satisfied without visiting, so the incentive to produce the content weakens, and the answering system depends entirely on that content continuing to exist.

Which is a genuine collective-action problem rather than a complaint about business models: the answering system is more useful per query and simultaneously degrades the corpus it requires, and no individual system can resolve that by behaving better, because the benefit of defecting accrues to whoever defects.

So the honest position is to state it — along with what you actually do about it: prominent attribution, links that are genuinely followed rather than decorative, and licensing arrangements where they apply (Module 12). Stating the problem and your response is the available honesty, and it is more than most systems offer.

- **Invest in retrieval first**, because **that is where the quality is (§1) and where the attention is not** — which makes it the highest-return work available.
- **Require and verify citations, and present the supporting passage (§2)** — the verification being the part that distinguishes a system with citations from a system with attribution.

- **Measure abstention deliberately**, by constructing questions your corpus cannot answer and recording what happens (§3).
- **Route by difficulty: answer directly where you are confident, and return a result list where you are not** — which is the design that uses both modes well rather than committing to one, and which requires a confidence estimate.
- **And state what you are not providing**, which is Module 13's subject. **Routing between answering and listing is the design most systems skip**, and it gets the benefit of both modes — the convenience of an answer where it is warranted, and the user's own judgement where it is not.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                          | What it covers for this module                                                                                               |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Lewis et al. — Retrieval-Augmented Generation (free)                              | <b>§1's architecture in the original</b> — read it against CSCE 638 Module 11, which covers the generation side.             |
| Rashkin et al. — Measuring Attribution in NLG (free)                              | <b>§2's requirement, formalised</b> — what it means for a citation to support a claim, and how to judge it.                  |
| Liu, Zhang & Liang — Evaluating Verifiability in Generative Search Engines (free) | <b>§2 and §3 measured on deployed systems</b> — and the attribution rates are sobering.                                      |
| Nakano et al. — WebGPT (free)                                                     | <b>§1's pipeline with retrieval in the loop</b> — and the honest discussion of what the citations did and did not establish. |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Build the answering pipeline** over your Project 2 collection.
- 2 **Measure passage-level retrieval recall** separately from answer quality.
- 3 **Generate per-sentence citations** and sample twenty for reading.
- 4 **Record the attribution rate** — the fraction genuinely supported.
- 5 **Instruct the model to quote first** and measure the attribution rate again.
- 6 **Add an entailment check** and report what it drops.
- 7 **Construct ten unanswerable questions** and measure the abstention rate.
- 8 **Reorder the passages worst-first** and measure the quality change.

- 9 **Compare answering against listing** on the same queries, by user judgement.
- 10 **Implement difficulty routing** between the two modes.

## Self-check

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Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What does returning an answer move, and what does the user lose?
- 2 How does the failure mode change?
- 3 Give the pipeline stages, and say which is the bottleneck.
- 4 Contrast weak and strong attribution.
- 5 What is the characteristic attribution failure, and why is it worse than no citation?
- 6 Give four ways to improve attribution, and the cheapest.
- 7 Name five things to measure, and the free signals.
- 8 State the consequence for document authors.
- 9 Why is it a collective-action problem?
- 10 What is the design most systems skip?